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Original Article

Natural resources and nuclear energy development: Does political stability matter?

Dianfei Luan [a,*], Feng Yang [b], Muhammad Hafeez [c,d]

[a] School of Culture and Tourism, Wuxi Vocational College of Science and Technology, Wuxi, Jiangsu, 214000, China
[b] Sports Center, Xi'an International University, Xi'an, Shaanxi, 710077, China
[c] Institute of Business Management Sciences, University of Agriculture, Faisalabad, 38040, Pakistan
[d] Adnan Kassar School of Business, Lebanese American University, Beirut, Lebanon

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ABSTRACT

Policymakers all around the globe are focusing on the factors that can contribute to ecological sustainability. Nuclear energy is considered a sustainable and low-carbon energy source. Therefore, finding the factors that can contribute to nuclear energy development is crucial. The analysis aims to estimate the influence of natural resource rents and political stability on nuclear energy development. To obtain the estimates of these variables, we employ the CS-ARDL and PMG-ARDL methods using panel data from 1996 to 2021. The results of these methods indicate that natural resource rents hurt nuclear energy development, while political stability contributes positively to increasing investment in nuclear energy in the long run only. The controlling factors, such as economic growth and trade, also hinder nuclear growth, and financial development is instrumental in fostering nuclear energy. The short-run estimates appear to be insignificant for most of the variables. In light of these findings, we suggest that policymakers should enhance the share of nuclear energy in the energy mix via the channels of political stability and natural resource rents.

1. Introduction

The start of the 19th century due to the Industrial Revolution brought a rapid paradigm change that drove the world's economy into an era of fast-paced economic growth [1]. As a result of the industrial revolution, the production cycle of the industries became automated, increasing the energy demand to achieve high-quality economic growth. A large number of economies depend on fossil-fuel-driven energy sources, including coal, oil, and gas, to fulfill the energy required by manufacturing activities and to initiate sustainable economic growth, which produce a significant amount of carbon footprints as byproducts and significantly alter the atmosphere [2]. It is clear that fossil fuels are the major factors behind economic development, but they also emit a lot of carbon and other toxic gases into the environment and enhance global temperature. Thus, the IPCC acknowledged in its fifth evaluation report that CO2 is the primary driver of temperature. In this regard, agreements like the Paris Agreement and UN sustainable development agenda stressed replacing fossil fuels with more clean and green energy sources, including nuclear and alternative energy sources, which are crucial for decarbonizing the economy without slowing economic growth [3].

Nuclear energy is the most valuable power source that produces no pollutants and spurs economic growth [4]. This is because nuclear reactors use the fission process to generate power. The method relies on the breakdown of uranium atoms to produce energy. In contrast to fossil fuels, this energy often produces steam that helps rotate the turbine to create electricity without producing toxic emissions. As a result, the USA avoided releasing 471 million metric tons of carbon only in 2021 [5]. The IEA [6] estimates that over the past five decades, nuclear energy has helped remove about 55 gigatons (Gt) of carbon equivalent from the atmosphere, comparable to pollutants from two years of continuous use of fossil fuels. Thus, there is an ongoing debate among economists, academics, and environmentalists regarding the benefits of nuclear for economic and environmental sustainability. Within this context, it is crucial to estimate the determinants of nuclear energy determinant, particularly the non-economic determinants of nuclear energy.

The nation's economic and political strength largely depends on natural resource extraction and consumption [7]. Natural resources continue to be a major revenue source for a large number of resource-rich nations. The majority of Middle Eastern nations rely on oil production for their economic and political dominance, whereas in the

* Corresponding author. School of Culture and Tourism, Wuxi Vocational of Science and Technology, Wuxi, Jiangsu 214000, China.
E-mail addresses: luandianfei_2016@163.com (D. Luan), yangyangdeyi768009@163.com (F. Yang), muhammad.Hafeez@uaf.edu.pk (M. Hafeez).

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developing and developed economies. The study observed the increasing impact of natural resources on RE, whereas the linkage is stronger in developed economies than in developing economies.

Few studies have examined the effect of natural resources on ecological pollution and green investment. For instance, Zhong & Kan [21] examined the ecological consequences of natural resources for China by employing the dynamic ARDL technique on a dataset spanning from 1981 to 2021. The study concluded that natural resources tend to reduce ecological degradation. Considering the data set from 1996 to 2021 for 51 green investing economies, Alsagr & Ozturk [12] study explored the linkage between natural resource rent and green investment. The study performed random effect and 2SLS estimation techniques and reported that natural resource rents are harmful for green investment.

Political stability is another important determinant of nuclear energy investment, but there is no empirical literature discussing the nexus between political stability and nuclear energy investment. However, few studies have investigated the impact of political stability on RE. In the case of MENA, Saadaoui [22] reported the significant influence of political stability on RE and energy efficiency policies. Abban & Hasan [23] examined the influence of government policy, government orientation, and nature of government system on RE investment in the case of 60 developed and developing economies. The study also considered parliamentary and presidential systems and possible types of ideology, namely left, right, and center in analysis. The study found that central and left-oriented ruling parties report stronger positive effect on RE investment than a right-orientated government. Additionally, the parliamentarian system is more effective for RE investment than the presidential system. The study also emphasized the importance of political stability in RE development.

Considering the dataset for the UK and Australia, Clulow et al. [24] explored the effect of public attitudes and political ideology on various energy technologies, including wind energy, wave, solar thermal, nuclear, hydroelectricity, carbon capture, carbon storage, natural gas, shale gas, coal and biomass energy. The findings report a supportive impact of political ideology on most energy technology variables including renewables and nuclear energy. Chen et al. [25] demonstrated the significant role of democratic institutions in stimulating RE. For European economies, Marques & Fuinhas [26] reported the negative effect of lobbying activities on RE deployment. On the other hand, Mehrara et al. [27] reported promoting the impact of institutional drivers, including political stability on RE in the case of ECO economies from 1992 to 2012. For the case of 26 European Union economies over time horizon 2004 to 2011, Cadoret & Padovano [28] investigated the nexus between political determinants and RE. The study found a positive effect of political stability on RE. Sequeira & Santos [29] also found a positive linkage between democracy and RE consumption.

The review of studies confirms that none of the studies in the existing literature have considered the role of natural resources and political stability in nuclear energy development. While numerous studies have considered the roles of natural resources and political stability in RE development, there is a lack of research on how these determinants impact nuclear energy. Existing studies have extensively addressed RE sources, leaving a gap in the context of nuclear energy. Thus, the present study examines the impact of natural resources and political stability, particularly on nuclear energy development.

3. Theoretical framework

The energy transition theory offers a theoretical explanation on the nexus between natural resources and nuclear energy. This theory posits that several factors are responsible for the shift to nuclear and alternative energy sources, including "economic, technological, political, and social" factors [30]. Thus, the nexus between natural resources and nuclear energy is not very straightforward and is a complex phenomenon explained via three mechanisms. First, economies with abundant natural resources need to increase the consumption of low-carbon energy sources to enhance their reputation at the global level [31]. Over the past few decades, ecological sustainability has gained a lot of attention. It has been linked to several aspects, such as tourism and other services, manufactured goods, socio-economic norms, and political authority. Hence, economies with abundant natural resources can't relinquish their commitments to achieve superior ecological performance as their legitimacy and morality on the global stage are compromised. The exploration and mining activities come with the hidden responsibility of using the natural resources in a moderate way that may not damage the ecosystem because these activities are the primary culprit behind the global climate change phenomenon. Hence, it is morally and socially obligatory for resource-abundant economies to increase their production of alternative and low-carbon energy sources. Second, nations with abundant natural resources have enough financial capital to produce alternative and nuclear energy. In order to produce and use nuclear and other low-carbon energies, significant capital expenditure is needed. In the case of nuclear and alternative energies, most of the original investment becomes sunk cost [9]. Thus, resource-rich nations are more likely to be able to afford significant investment in nuclear energy and alternative energy sources. Third, the future belongs to low-carbon energy sources. Thus, resource-rich nations must invest in low-carbon energy initiatives using their previous energy production experience to keep themselves relevant in the energy markets.

The political economics of energy and ecology is primarily explained by institutional quality, especially the government's political ideology [28]. There is a direct connection between investment in nuclear energy and the government's ideology and the efficiency of its institutions. According to Fredriksson & Svensson [32], environmental restrictions weaken due to increased political unrest and substantial corruption, which may discourage investment in nuclear and alternative energy projects. This is due to the fact that an uncertain political climate may significantly enhance economic uncertainty, making investment in nuclear and alternative energy sources riskier [33]. The increased violence in society due to unstable political conditions may lead to a weakening state of law and order in the community. Consequently, the government has to divert resources from economic activities toward the prevention of violence, which has a detrimental effect on public support for research and development. Therefore, political turmoil may have a negative impact on the development of renewable energy technologies [34]. On the other hand, governments in a politically stable environment are in a better position to earmark funds for investment in nuclear and RE projects. Further, the government's energy policies within a stable political framework are much more consistent, which is almost a prerequisite for investment in nuclear energy projects. Based on the above theoretical framework, we have developed the research hypotheses.

H1. More natural resources lead to higher nuclear energy development.

H2. Political stability increases nuclear energy development.

4. Model and methodology

4.1. Econometric model

This study follows the empirical models model [9,34], which links nuclear energy development, natural resources, and political stability. The econometric model postulates that natural resources, political stability, economic growth, financial development, technology, and trade drive long-run nuclear energy. Therefore, we model it as:

$$NED_{it} = \partial_0 + \partial_1 NR_{it} + \partial_2 PS_{it} + \partial_3 EG_{it} + \partial_4 FD_{it} + \partial_5 TECH_{it} + \partial_6 TRADE_{it} + e_{it} \quad (1)$$

where nuclear energy development (*NED*) is a function of natural

resources (NR), political stability (PS), economic growth (EG), financial development (FD), technology (TECH), trade openness (TRADE), and error term (e_{it}). Natural resource rents can exert both positive and negative affect on nuclear energy development. On the positive side, resource-rich economies use the revenues generated from natural resource extraction to invest in nuclear energy infrastructure. Conversely, reliance on natural resource rents can negatively influence nuclear energy development by creating an economic dependence on fossil fuels. Political stability is expected to have a positive effect on nuclear energy development. Political stability plays a crucial role in enhancing nuclear energy development, as it provides a foundation for sustained economic growth.

4.2. Econometric methodology

4.2.1. Cross-sectional dependence and homogeneity tests

Globalization and economic integration have greatly enhanced interdependence between the economies, making cross-sectional dependence (CSD) a persistent characteristic of panel data analysis. The first step of panel data analysis involves assessing the existence of a relationship among cross-sectional units. Therefore, before determining the impact of natural resources and political stability on nuclear energy development, it is essential to analyze CSD. To do so, we selected the Pesaran CSD test [35]. The test assesses the null hypothesis of cross-sectional independence in contrast to the alternative hypothesis of cross-sectional dependency. The CSD test statistics are based on the following equation:

$$CSD_{test} = \sqrt{\frac{2T}{N(N-1)}}\left(\sum_{i=1}^{N-1}\sum_{k=i+1}^{N}\widehat{\tau}_{ik}\right) \tag{2}$$

Checking the heterogeneity in slope across cross-sections, together with CSD, is a second preliminary test in a panel data analysis. This study focused on the Pesaran & Yamagata [36] test to assess the homogeneity or heterogeneity of the slope. The following equations evaluate slope coefficients:

$$\widetilde{\Delta}_{HPY} = (N)^{\frac{1}{2}}(2k)^{-\frac{1}{2}}\left(\frac{1}{N}\widetilde{S} - k\right) \tag{3}$$

$$\widetilde{\Delta}_{ASH} = (N)^{\frac{1}{2}}\left(2k\left(\frac{T-k-1}{T+1}\right)^{-\frac{1}{2}}\left(\frac{1}{N}\widetilde{S} - 2k\right)\right) \tag{4}$$

4.2.2. Panel unit root tests

Thirdly, understanding the stationary properties of the variables is essential for selecting appropriate estimating methods and, therefore, obtaining accurate estimates. For this purpose, two generations of unit root tests are available. First-generation panel unit root tests (LLC, IPS, ADF, and PP) fail to perform accurate results because they ignore CSD. The second-generation tests more effectively address CSD and slope heterogeneity, so we choose the CIPS test [37]. The equation given below serves as the basis for this test:

$$\Delta V_{i,t} = \alpha_i + \alpha_i X_{i,t-1} + \alpha_i Y_{t-1} + \sum_{l-0}^{p}\alpha_{i1}\Delta\overline{V_{t-1}} + \sum_{i-1}^{p}\alpha_i\Delta Y_{i,t-1} + \mu_{it} \tag{5}$$

Specification (5) depicted the cross-sectional averages. The CADF method is extracted from specification (5), which ultimately produces the CIPS statistics as shown below:

$$\widehat{CIPS} = \frac{1}{N}\sum_{i=1}^{n} C_{ADF}^{i} \tag{6}$$

4.2.3. Panel cointegartion tests

Additionally, panel cointegration tests, using the approach established by Westerlund [38], are used to analyze the long-term equilibrium relationships between natural resources, political stability, and nuclear energy development. These tests ascertain if variables simultaneously move over the long term, indicating reliable long-term connections crucial for policy suggestions. This research favours the panel cointegration test of Westerlund, which is widely used because of its ability to do well in the presence of CSD. The following equation (7) describes the error-correction framework:

$$\Delta x_{it} = \sigma_i d_t + \delta_i\left(x_{it-1} - \beta_{i\ y_{it-1}}\right) + \sum_{j=1}^{ki}\delta_{ij}\Delta x_{it-j} + \sum_{j=-qi}^{ki}\gamma_{ij}\Delta y_{it-j} + \varepsilon_{it} \tag{7}$$

where the adjustment term is signified by δi in Specification (7), it indicates the rate at which all variables achieve equilibrium in the long-run. The least-squares estimates lay the foundation for this test δi, whereby the null hypothesis supports the non-existence of cointegration. Equations (8) and (9) are used for estimating the group mean statistics, while equations (10) and (11) represent the panel statistics:

$$G_\tau = N^{-1}\sum_{i=1}^{N}\frac{\theta_i}{SE(\theta_i)} \tag{8}$$

$$G_\alpha = N^{-1}\sum_{i=1}^{N}\frac{T\theta_i}{\theta_i(1)} \tag{9}$$

$$P_\tau = \frac{\theta_i}{SE(\theta_i)} \tag{10}$$

$$P_\alpha = T\theta_i \tag{11}$$

One panel and group statistic should be significant to reject the null hypothesis or prove cointegration.

4.2.4. CS-ARDL method

Based on the preliminary test findings, particularly the presence of CSD and heterogenous slope coefficients, we picked Chudik & Pesaran's [39] CS-ARDL method. This is crucial for estimating the short and long-run connection between natural resources, political stability, and nuclear energy development. This is widely considered superior to several other techniques used for panel analysis. Among the several advantages of the CS-ARDL, producing short and long-run estimates simultaneously, controlling CSD and endogeneity, employing a mean group technique with changing slope coefficients, and allowing the inclusion of variables in the model with varying order of integration (I(0) and I(1)), are worth mentioning. This approach relies on the mean group; thus, it is considered an advanced remodeling of the ARDL framework. Therefore, for every panel entity, in order to address the possible bias caused by the unobserved factors, the CS-ARDL estimates the cross-sectional mean for each of these factors and the lags attached to them. Further, the insertion of the lagged value of the cross-sectional means within the model is crucial for resolving the endogeneity issue. The specification used to represent the CS-ARDL model is given below:

$$\Delta NED_{it} = C_i + \Psi_i(NED_{it-1} - \Upsilon_i Z_{it-1} - \phi_{1i}\overline{NED}_{t-1} - \phi_2\overline{Z}_{t-1}) + \sum_{j=1}^{p-1}\omega_{ij}\Delta NED_{it-j} + \sum_{j=0}^{q-1}\lambda_{ij}\Delta Z_{it-j} + \pi_{1i}\Delta\overline{NED}_t + \pi_{2i}\Delta\overline{Z}_t + e_{it} \tag{12}$$

The outcome variable in equation (12) is nuclear energy development (ΔNED_{it}). This variable is a function of several regressors, including natural resources, political stability, financial development, technology, and trade openness, in the LR, symbolized by Zit. In addition, NED_{t-1} and Z_{t-1} signify the averages of the outcome variable and regressors in the long-run; however, in the short-run the mean values are represented by ΔNED_t and ΔZ_t, respectively. Lastly, ε_{it} is a white noise

residual term.

We also employ the PMG-ARDL technique [40] for robustness check due to its similarity with CS-ARDL in many aspects. For instance, the ARDL is optimal for assessing outcomes when the data series exhibits mixed-order stationarity. The ARDL-PMG is favored for its capacity to evaluate both short- and long-term effects and the cross-sectional short-term impacts of the logical variables on the ward. Thus, it also fulfills our objective of producing short and long-run estimates. However, it can't account for the issue of CSD, and to counter that, we additionally employ the FMOLS and DCCE methods.

5. Data and descriptive analysis

This study investigates the effect of natural resources and political stability on nuclear energy development using top 17 nuclear energy-producing economies[1] from 1996 to 2021. The nuclear energy development (NED) data series is collected from the Energy Information Administration (EIA) and is measured in terms of total energy production from nuclear sources in quad Btu. Independent variables are natural resources (NR) and political stability (PS). Natural resources (NR) are assessed from total natural resources rents as percent of GDP. Whereas, political stability (PS) is measured by the estimates of political stability and absence of violence and terrorism. The data series for NR and PS are assembled from the World Development Indicators (WDI) and Worldwide Governance Indicators (WGI), respectively. Economic growth (EG), financial development (FD), technology (TECH), and trade openness TRADE) are control variables in our model. The data series for economic growth, technology, and trade openness are sourced from the WDI, while financial development data is collected from the International Monetary Fund (IMF). Economic growth stimulates demand for energy as industries and infrastructure expand. A higher GDP growth provides financial capacity to support the expensive technology and safety measures associated with nuclear energy development [41]. In our analysis, economic growth is assessed from annual percent of GDP growth. Financial development is included because a well-developed financial sector facilitates the mobilization of funds for nuclear energy projects [42]. An index is used to measure financial development. The third control variable is technology, which plays a crucial role in enhancing nuclear energy development [43]. Technology (TECH) variable is determined through total patent applications. Lastly, trade can lead to cumulative gains that reduce costs and accelerate energy transitions [44]. Total trade as percent of GDP is used to measure trade openness.

In Table 1, the descriptive statistics reveal that NED has a mean of 1.166 quad Btu, with a S.D of 2.035, with values ranging from 0.000 to 8.459. NR has a mean of 1.920 and a S.D of 1.998. The minimum value is 0.012, and the maximum is 11.99. PS shows a mean of 0.081 and a S.D of 0.894, with values ranging from −2.810 to 1.759. EG has a mean of 2.587, with a S.D of 3.795. The minimum value of −15.13 highlights significant economic downturns in some countries, while the maximum value of 14.23 reflects periods of growth. FD has a mean value of 0.596 and a S.D of 0.236, with values ranging from 0.092 to 0.947. TECH has a mean of 9.470 and a S.D of 1.942, with values ranging from 6.425 to 14.28. Lastly, TRADE has a mean of 3.977 and a S.D of 0.582. The values range from 2.750 to 5.151.

The correlation matrix in Table 2 presents the pairwise relationships between variables. NED is positively correlated with FD (0.441), TECH (0.576), and PS (0.215). However, NED is negatively correlated with NR (−0.270), EG (−0.139), and TRADE (−0.305). NR shows a negative correlation with PS (−0.419), FD (−0.408), and TRADE (−0.090), but a positive correlation with EG (0.258) and TECH (0.059). PS has strong positive correlations with FD (0.584), TRADE (0.344), and TECH (0.140). PS is negatively correlated with EG (−0.194). The results of Table 2 concluded that none of the variables exhibit perfect correlation, indicating that we can proceed with the regression analysis.

The VIF test, presented in Table 3, is used to examine the presence of potential multicollinearity among the variables. The VIF scores are below 10 for all variables, suggesting that there is no evidence of multicollinearity. The highest VIF is observed for TECH at 2.41, followed closely by FD at 2.34, and PS at 2.04. TRADE and NR have lower VIF values of 1.98 and 1.51, respectively, while EG exhibits the lowest VIF at 1.10, indicating very low multicollinearity for this variable. The mean VIF for all variables is 1.89, further confirming the absence of multicollinearity among independent variables.

6. Empirical results

Before CS-ARDL analysis, CSD and slope homogeneity are tested. The study uses Pesaran test to check for CSD. Table 4 presents the results of the CSD test. The significant values for most variables (NED, NR, PS, EG, FD, and TRADE) suggest the presence of CSD, except for the variable TECH. Table 5 presents the results of the slope homogeneity test for all variables. The results indicate that NED, NR, PS, FD, TECH, and TRADE all have significant values at the 1 % level for both Delta and $Delta_{adj}$ statistics, suggesting slope heterogeneity across the sample. In contrast, EG shows insignificant results for both tests, implying slope homogeneity. The findings of the CSD and slope homogeneity tests also imply that the CIPS unit root test should be applied. The CIPS test assesses the stationarity of the variables at both levels (I(0)) and first differences (I(1)). Table 6 shows the estimates of CIPS test. The results indicate that NR, PS, EG, and FD are I(0) stationary. In contrast, NED, TECH, and TRADE are non-stationary at the level, but they become stationary after differencing. NED, TECH, and TRADE all have significant results at first difference (I(1)). Table 7 presents the results of the cointegration test. The results validate the presence of a LR relationship between the variables.

The LR and SR results of CS-ARDL and PMG-ARDL are demonstrated in Table 8. In the LR, a significantly negative linkage between NR and NED exists, which shows that for every 1 % rise in NR, NED will decline by 1.235 % and 0.024 % in both models. By contrast, PS positively correlate with NED in both models, but it is insignificant in CS-ARDL. Correspondingly, NED significantly rises by 0.175 % for 1 % rise in political stability in the PMG-ARDL. EG is significantly and negatively linked with NED in both models. The LR coefficients of the EG are 0.028 and 0.012 which means that NED will decline by about 0.028 % and 0.012 % for every 1 % increase in the EG in the LR. The LR estimates of FD are statistically significant and positive in both models. The LR coefficients of FD are 1.062 and 0.371 meaning that NED will increase by almost 1.062 % and 0.371 % for every 1 % rise in FD in the LR. The LR estimates of TECH are positive in both models, but the coefficients are statistically insignificant. TRADE is significantly and negatively linked with NED in both models. The LR coefficients of the TRADE are 0.160 and 0.415 which means that NED will decline by almost 0.160 % and 0.415 % for every 1 % increase in the TRADE in the LR. The summary of key results is presented in Fig. 2.

The SR findings illustrate that D(NR) elevates NED positively, but the estimates are insignificant. Moreover, the D(PS) also stimulates NED, but the coefficients are insignificant in the SR. By contrast, D(EG) significantly reduces NED in both models in the SR. The estimates infer that due to a 1 % rise in D(EG), NED declines by 0.015 % in the CS-ARDL and 0.005 % in the PMG-ARDL in the SR. The SR effect of D(FD) on NED is also negative but insignificant. The SR coefficients of D(TECH) are positive in both models but the effect is only significant in the PMG-ARDL. It means that a 1 % upsurge in D(TECH) enhances NED by 0.262 % in the PMG-ARDL in the SR. Although the SR effect of D(TRADE) on NED is positive in both models, but estimates are statistically insignificant. In the end, ECM (−1) terms are significant and negative in both models, confirming the tendency to achieve stability in

[1] China, USA, Japan, India, Canada, South Africa, Mexico, Brazil, UK, Spain, France, Pakistan, Argentina, Ukraine, Czechia, Netherlands, Belgium.

Table 1
Definitions and sources.

Variables	Definitions	Sources	Mean	Std. dev.	Min	Max
Nuclear energy development (NED)	Total energy production from nuclear (quad Btu)	EIA	1.166	2.035	0.000	8.459
Natural resources (NR)	Total natural resources rents (% of GDP)	WDI	1.920	1.998	0.012	11.99
Political stability (PS)	Political stability and absence of violence/terrorism: estimate	WGI	0.081	0.894	−2.810	1.759
Economic growth (EG)	GDP growth (annual %)	WDI	2.587	3.795	−15.13	14.23
Financial development (FD)	Financial development index	IMF	0.596	0.236	0.092	0.947
Technology (TECH)	Patent applications, total	WDI	9.470	1.942	6.425	14.28
Trade openness (TRADE)	Trade (% of GDP)	WDI	3.977	0.582	2.750	5.151

Source: Authors calculation.

Table 2
Correlation matrix.

	NED	NR	PS	EG	FD	TECH	TRADE
NED	1						
NR	−0.270	1					
PS	0.215	−0.419	1				
EG	−0.139	0.258	−0.194	1			
FD	0.441	−0.408	0.584	−0.081	1		
TECH	0.576	0.059	0.140	0.070	0.483	1	
TRADE	−0.305	−0.090	0.344	−0.047	0.002	−0.531	1

Source: Authors calculation.

Table 3
VIF results.

Variable	VIF	1/VIF
TECH	2.41	0.416
FD	2.34	0.428
PS	2.04	0.490
TRADE	1.98	0.506
NR	1.51	0.661
EG	1.10	0.912
Mean VIF	1.89	

Source: Authors calculation.

Table 6
CIPS test results.

	I(0)	I(1)
NED	−1.084	−4.456***
NR	−2.069***	
PS	−1.871**	
EG	−3.493***	
FD	−2.183***	
TECH	−1.765	−4.858***
TRADE	−1.387	−4.020***

Note: ***p < 0.01, **p < 0.05, *p < 0.1.

the LR. The ECM (−1) values are −0.871 and −0.347, suggesting that 87 % of disequilibrium will be corrected within one year in the CS-ARDL, while 35 % will be adjusted over the same period in the PMG-ARDL.

The FMOLS and DCCE results in Table 9 assess the robustness of CS-ARDL and PMG-ARDL results. The estimations reveal that NR have a significant negative impact on NED in both models. Specifically, the FMOLS estimation yields a coefficient of −3.162, while the DCCE estimation reports a coefficient of −1.592. This means that a 1 % rise in NR reduces NED by 3.162 % and 1.592 % in the FMOLS and DCCE. PS exhibits a positive and significant effect on NED across both models, with coefficients of 0.245 in FMOLS and 0.122 in DCCE. It depicts that a 1 % rise in PS tends to expand NED by 0.245 % in the FMOLS and 0.122 % in the DCCE. EG shows a slight negative influence that is statistically significant only in the FMOLS, with a coefficient of −0.016; it is not significant in the DCCE. FD is positive and significant in both models. The FMOLS coefficient is 0.672, and the DCCE coefficient is 0.552. TECH positively affects the dependent variable in both estimations. The FMOLS presents a coefficient of 0.265, while the DCCE shows a coefficient of 0.393. TRADE has a significant negative effect in the FMOLS

Table 7
Cointegration test results.

Statistic	Value	z-value	P-value
Gt	−1.734***	3.003	0.001
Ga	−6.117**	2.098	0.018
Pt	−5.518***	2.924	0.002
Pa	−4.538***	5.001	0.000

Note: ***p < 0.01, **p < 0.05, *p < 0.1.

Table 4
CSD test results.

	NED	NR	PS	EG	FD	TECH	TRADE
Pesaran's test	4.694***	21.47***	4.267***	27.78***	10.52***	−0.683	6.430***
Off-diagonal elements	0.365	0.427	0.310	0.504	0.459	0.381	0.350

Note: ***p < 0.01, **p < 0.05, *p < 0.1.

Table 5
Slope homogeneity test results.

	NED	NR	PS	EG	FD	TECH	TRADE
Delta	8.002***	7.833***	6.074***	1.025	13.59***	11.80***	12.73***
$Delta_{adj}$	9.617***	9.414***	7.300***	1.232	16.33***	14.18***	15.31***

Note: ***p < 0.01, **p < 0.05, *p < 0.1.

Table 8
CS-ARDL and PMG-ARDL results.

	CS-ARDL	PMG-ARDL
Long-run		
NR	−1.235**	−0.024**
	(0.533)	(0.010)
PS	0.087	0.175***
	(0.121)	(0.033)
EG	−0.028*	−0.012***
	(0.014)	(0.003)
FD	1.062**	0.371**
	(0.518)	(0.145)
TECH	0.138	0.024
	(0.163)	(0.032)
TRADE	−0.160	−0.415****
	(0.354)	(0.102)
Short-run		
D(NR)	1.084	0.270
	(0.951)	(0.450)
D(PS)	0.088	0.059
	(0.096)	(0.045)
D(EG)	−0.015*	−0.005**
	(0.008)	(0.002)
D(FD)	−0.794	−0.535
	(0.818)	(0.360)
D(TECH)	0.023	0.262**
	(0.134)	(0.122)
D(TRADE)	0.085	0.198
	(0.159)	(0.177)
C		0.015
		(0.127)
Observations	423	423
Number of countries	17	17
ECM(-1)	−0.871***	−0.347***
	(0.101)	(0.104)

Note: ***p < 0.01, **p < 0.05, *p < 0.1.

Table 9
FMOLS and DCCE results (robustness).

Variables	FMOLS	DCCE
NR	−3.162***	−1.592**
	(0.424)	(0.710)
PS	0.245***	0.122*
	(0.033)	(0.069)
EG	−0.016***	−0.005
	(0.002)	(0.016)
FD	0.672***	0.552**
	(0.197)	(0.269)
TECH	0.265***	0.393*
	(0.033)	(0.217)
TRADE	−0.534***	−0.178
	(0.111)	(0.288)
Constant	1.420***	
	(0.543)	
R-squared	0.467	0.328
Number of counties	17	17

Note: ***p < 0.01, **p < 0.05, *p < 0.1.

estimation, with a coefficient of −0.534, but its effect is not statistically significant in the DCCE.

6.1. Results discussion

This study's first and foremost result reveals an unfavorable relationship between natural resource rents and nuclear energy deployment. This result is against the hypothesis 1. Resource-rich economies are more unlikely to invest in nuclear and alternative energy projects. This finding is also supported by Shinwari et al. [45], who noted that natural resources negatively impact renewable energy investment. The intuition behind this negative connection could be that resource-rich economies are not taking the global pressure with regard to increasing investment in low-carbon energy projects. Moreover, they rely too much on natural resource extraction and mining activities and ignore the development of other sectors, including energy, thus hindering the deployment of nuclear and alternative power projects. This finding is also backed by Fan et al. [46], who state that the ecosystem is severely damaged during the mining of natural resources. However, some studies oppose our findings by confirming the positive role of natural resource rents in fostering the development of alternative and low-carbon energy sources [31].

The second significant outcome of the analysis is the positive link between political stability and nuclear energy deployment. This finding is in line with hypothesis 2. This is justified because a nation's economic performance largely relies on its political stability [47], and it has become instrumental in fostering the sustainable development of the economy's each sector, and the energy sector is no exception. In order to develop a conducive environment for investment in nuclear power plants, political stability has a crucial role. It helps overcome the risks related to investments in nuclear energy and thus attracts more FDI for the nuclear sector. Dincer & Uluer [48] expressed that socio-political aspects are essential in attracting financial investment for nuclear energy initiatives. Investors want public endorsement and governmental

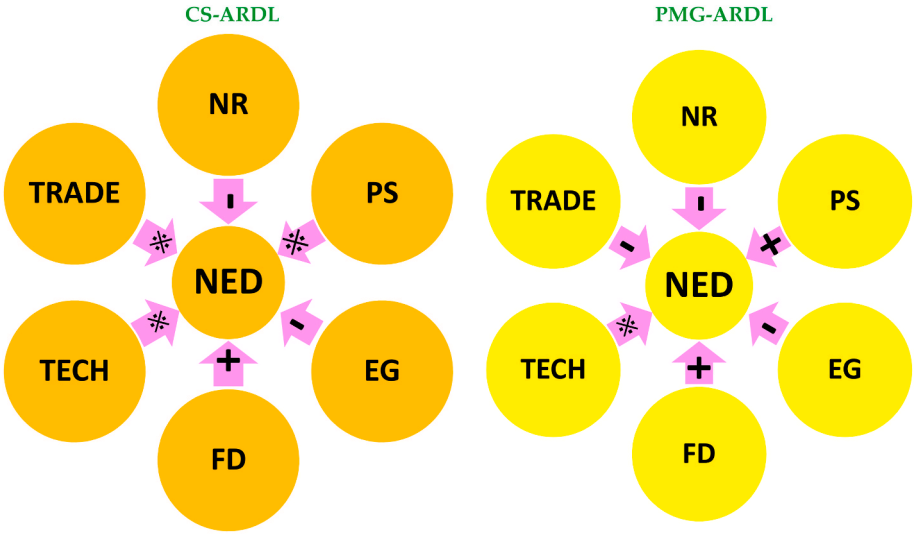

Fig. 2. Summary of key results. Note: +: positive effect; −: negative effect; ※: No effect.

dependability for investments in high-risk nuclear energy initiatives with long payback periods and low-profit rates. They further noted that a stable political climate stimulates investment in nuclear energy by enabling the development of technological infrastructure and building a harmonious environment on the pros of constructing nuclear power facilities. Likewise, Jewell [33] also posited that political stability facilitates the effectiveness of governmental operations and promotes technical advancement, which is crucial for the deployment of nuclear and alternative energy projects. These results are also supported by the findings of Wang et al. [34], Omri & Jabeur [49], and Dinçer & Uluer [48].

7. Conclusions and implications

Policymakers all around the globe are focusing on the factors that can contribute to ecological sustainability. In this regard, the role of low-carbon energy sources, including nuclear energy, has received worldwide recognition as a contributor to ecological sustainability. Nuclear energy is considered a sustainable and low-carbon energy source; however, there is controversy regarding the development of nuclear energy. Therefore, finding the factors that can contribute to nuclear energy development is crucial. The literature on the determinants of nuclear energy is limited, particularly since no past study has combined economic and non-economic factors in nuclear energy function, creating a big loophole in the literature. This analysis fills the gap in the literature by analyzing the influence of natural resource rents and political stability on nuclear energy development. To obtain the estimates of these variables, we employ the CS-ARDL and PMG-ARDL frameworks. The results of these methods indicate that natural resource rents hurt nuclear energy development, while political stability contributes positively to increasing investment in nuclear energy in the LR only. The controlling factors, such as economic growth and trade, also hinder nuclear growth, and financial development is instrumental in fostering nuclear energy. The SR estimates appear to be insignificant for most of the variables.

In light of these findings, the following policy recommendations are presented. Natural resource rents and nuclear energy are negatively connected, suggesting that policymakers focus more on extraction and mining activities and rely on conventional energy sources. Thus, a serious effort is required on behalf of the policymakers to divert their focus from conventional to alternative energy sources. In this regard, it is suggested that resource rents should be used to foster research and development activities in the energy sector, which pave the way for the development of alternative energy sources. Government should allocate a portion of resource rent for the investment in nuclear energy power projects and other alternative energy resources. The government should redesign the state policy to rely on the natural resource sector and energy production by using natural resources. A well-balanced energy policy is required, slowly but surely replacing fossil fuel energy sources with clean and green ones. Moreover, political stability is essential for nuclear energy deployment, indicating that policymakers must implement significant measures to stabilize the political landscape. To maintain and protect the continuous development of nuclear energy during periods of political instability, governments must avoid reallocating public funds from innovation to preserve law and order. Additionally, economic and political instability discourages investment. Countries experiencing political instability must employ forward-thinking strategies to attract investment and speed up economic recovery.

This study has made significant contributions to the literature; nonetheless, this work has some limitations that provide possible opportunities for further research. Contemporary works suggest that non-linear modelling is better suited for macroeconomic variables, especially during business cycles. This is significant because macroeconomic series, such as natural resources and politics, are influenced by human behaviour and exhibit asymmetrical characteristics. Thus, we recommend that future empirical studies use non-linear modelling to estimate the above-stated nexus. Furthermore, our research selected 17 top nuclear energy economies, including advanced and emerging economies. A comparative study between these two groups of countries is lacking, which could clarify how nuclear energy production responds to varying levels of political stability and natural resources across countries with distinct political, social, and economic frameworks. Consequently, we recommend conducting a comparative study between developed and developing countries.

CRediT authorship contribution statement

Dianfei Luan: Funding acquisition, Methodology, Writing – original draft, Writing – review & editing. **Feng Yang:** Data curation, Funding acquisition, Visualization, Writing – original draft, Writing – review & editing. **Muhammad Hafeez:** Writing – review & editing, Writing – original draft, Software, Investigation, Formal analysis, Conceptualization.

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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